FIRST SEMESTER B.Sc. DEGREE EXAMINATION, JANUARY 2014

(UG—CCSS)

Complementary Course

Statistics

ST IC 01—PROBABILITY THEORY

Time : Three Hours Maximum : 30 Weightage

Part A

Answer all **twelve** *questions.*
A bunch of **four** *questions carries weightage* 1.

1. If A and B are events which have no points in common, then the events A and B are :

   (a) Equally likely. (b) Independent.

   (c) Mutually exclusive. (d) Dependent.

2. If A and B are independent events such that $P(A) = P(B) = \frac{1}{2}$, then $P(A \cup B)$ is :

   (a) $\frac{1}{2}$. (b) 3/4.

   (c) $\frac{1}{4}$. (d) 1.

3. Which of the following is not an axiom of probability ?

   (a) $P(A) \geq 0$.

   (b) $P(S) = 1$.

   (c) $P(\phi) = 0$.

   (d) $P(A \cup B) = P(A) + P(B)$ when $A \cap B = \phi$.

4. If A and B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A/B) = \frac{1}{4}$, then $P(AB)$ is :

   (a) $\frac{1}{3}$. (b) $\frac{1}{8}$.

   (c) $\frac{1}{2}$. (d) $\frac{1}{24}$.

5. If A and B are independent events, then :

(a) A^c and B are independent. (b) A and $(A \cup B)$ are independent.

(c) A and $(A \cap B)$ are independent. (d) A and $\left(A^c \cup B\right)$ are independent.

6. If $F(x)$ is the d.f. of a random variable, then $F(+\infty) - F(-\infty)$ is :

(a) 0. (b) $\frac{1}{2}$.

(c) $+\infty$. (d) 1.

7. If $A \subset B$, then which of the following is true :

(a) $P(A) \leq P(B)$. (b) $P(A) < P(B)$.

(c) $P(A) > P(B)$. (d) $P(A) \geq P(B)$.

8. Skewness is a measure of :

(a) Spread. (b) Concentration.

(c) Symmetry. (d) Peakedness.

9. $E\left(e^{itX}\right)$ is known as :

(a) Moment generating function. (b) p.m.f.

(c) p.d.f. (d) Characteristic function.

10. If $M_X(t)$ is the MGF of X, then MGF of $Y = X + 3$ is :

(a) $e^{3t} M_X(t)$. (b) $M_X(3t)$.

(c) $3 + M_X(t)$. (d) $3M_X(t)$.